SDS 320E – Preliminary Analysis Report

Write in complete sentences using single-spaced, 12-pt font. Include figures in text. ***3-page limit.***

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Introduction

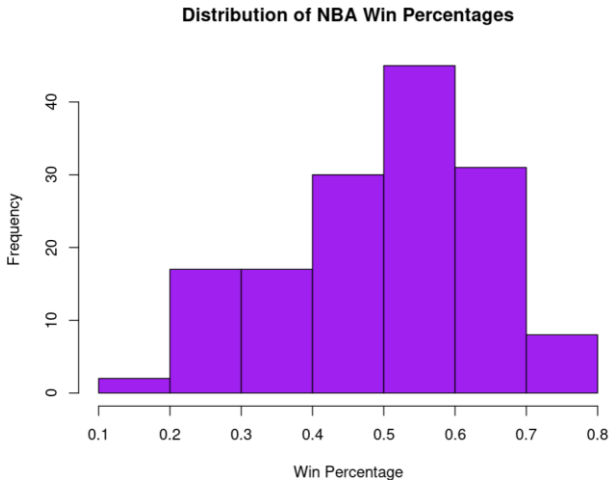
The goal of this study is to evaluate how three-point shooting behavior and efficiency relate to overall team success in the NBA. This data was collected from the Basketball Reference website (https://www.basketball-reference.com), which provided free access to extensive NBA statistics, including team-level data on three-point attempts, three-point efficiency, and win percentage over multiple seasons. The sampling units are NBA teams, with each observation representing one team's statistics for a single season. The final sample contains 150 observations, spanning from the 2019-2020 season to the 2023-2024 season.

Research Questions

RQ1 – Does the number of three-pointers attempted per game correlate with a team's win percentage?

RQ2 – Does a higher three-point shooting percentage lead to a higher win percentage?

Descriptive Analysis of Response Variable *(include graph of distribution here)*

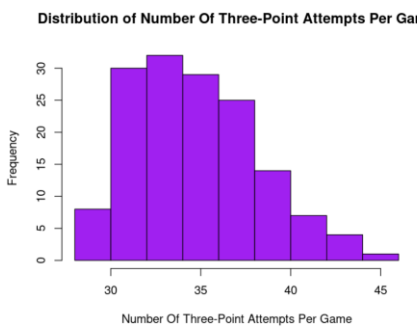


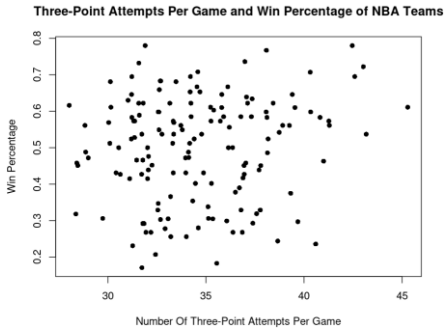
The histogram of win percentages appears to be slightly left-skewed. The median of this distribution is 0.524, meaning that 50% of the teams across these five seasons have won at least 52.4% of their games. The interquartile range (IQR) is 0.208, indicating that the middle 50% of teams had win percentages between 0.402 and 0.610.

Investigation of Numeric Explanatory Variable *(include univariate, bivariate graph here)*

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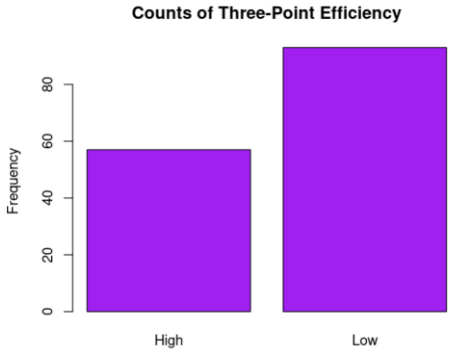


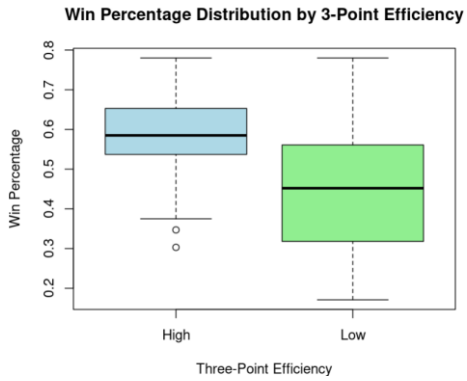


The histogram of the average number of three-point attempts per game appears to be slightly right-skewed. The median of this distribution is 34.2005 threes, meaning that 50% of the teams across these five seasons have shot at 34.2005 threes per game. The interquartile range (IQR) is 5.012 threes, indicating that the middle 50% of teams attempted between 31.9390 and 36.9510 threes per game.

Based on the scatterplot, there seems to be a very weak positive relationship between the number of three-point attempts per game and win percentage.

<u>Investigation of Categorical Explanatory Variable</u> *(include univariate, bivariate graph here)*





Among the 150 observations, 57 teams (about 38%) were classified as high efficiency, while 93 teams (about 62%) were classified as low efficiency.

Based on the boxplot, teams with high three-point efficiency appear to have higher win percentages on average compared to teams with low efficiency.

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R Code *(organized by variable without output or extraneous syntax)*

```
> nba_data <- read_csv("NBA Data Collection.csv")
> hist(nba_data$win_pct, xlab='Win Percentage', main='Distribution of NBA Win Percentages', right=F, col='purple', breaks=7)
> fivenum(nba_data$win_pct)
> hist(nba_data$threes_attempted, xlab='Number Of Three-Point Attempts Per Game', main='Distribution of Number Of Three-Point Attempts Per Game', right=F, col='purple', breaks=7)
> fivenum(nba_data$threes_attempted)
> plot(nba_data$threes_attempted, nba_data$win_pct, xlab='Number Of Three-Point Attempts Per Game', ylab='Win Percentage', main='Three-Point Attempts Per Game and Win Percentage of NBA Teams', pch=16)
> barplot(table(nba_data$threes_efficiency), ylab='Frequency', main='Counts of Three-Point Efficiency', col='purple')
> table(nba_data$threes_efficiency)
> prop.table(table(nba_data$threes_efficiency))
> boxplot(win_pct~threes_efficiency, data=nba_data, xlab='Three-Point Efficiency', ylab='Win Percentage', main='Win Percentage Distribution by 3-Point Efficiency', col=c('light blue','light green'))
```